Documentation of the System Development Methodology and Approaches

Project Title: Development of Integrated E- Library Management System

Client Organization: Ethiopian Geological Institute (EGI)

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1. Introduction

The Ethiopian Geological Institute manages large volumes of geological reports, GIS datasets, maps, laboratory records, research documents, and administrative data. Currently, information is stored in spreadsheets, printed files, local computers, and separate repositories, leading to limited access, weak collaboration, slow retrieval, and inconsistent metadata management.

To address these issues, EGI plans to develop an Integrated Geological Information Management System that supports centralized data management, advanced search, GIS integration, online access, digital preservation, and controlled distribution of geological resources.

This document provides the proposed methodology, system architecture, database structure, metadata framework, workflows, and implementation approach for a secure and scalable web-based system.

2. System Development Methodology

2.1 Agile Scrum Methodology

The project will use Agile Scrum methodology for flexible and iterative development. This approach supports continuous stakeholder involvement, rapid feedback, and gradual delivery of system features.

It also enables EGI stakeholders to review progress regularly before final deployment.

Benefits

- Continuous stakeholder engagement
- Faster issue resolution
- Incremental feature delivery
- Improved flexibility and scalability
- Early MVP development
- Reduced project risks
- Better alignment with institutional needs

2.2 Development Phases

Phase 1: Requirement Analysis

Activities include:

- Stakeholder interviews
- Site assessments
- Existing system evaluation
- Requirement gathering
- Workflow analysis
- Documentation review

Phase 2: System Design

Activities include:

- Database schema design
- Metadata structure preparation
- System architecture design
- API structure planning
- User interface prototyping
- Security framework design

Phase 3: System Development

Activities include:

- Frontend development
- Back-end API implementation
- Database development
- Elasticsearch integration
- Payment gateway integration

Phase 4: Testing and Quality Assurance

Activities include:

- Unit testing
- Integration testing
- Security testing
- Performance testing
- User Acceptance Testing (UAT)

Phase 5: Deployment and Training

Activities include:

- Production deployment

- Data migration
- User training
- Technical documentation
- System handover

3. Proposed System Architecture

3.1 Three-Tier Architecture

The proposed system will follow a three-tier architecture model composed of Presentation Layer, Application Layer, and Data Layer.

3.2 Presentation Layer (Front-end)

The Presentation Layer provides user interaction interfaces and dashboards.

Technologies

- React.js
- HTML and CSS
- JavaScript

Key Features

- Responsive dashboard
- User-friendly navigation
- Mobile accessibility
- Interactive GIS visualization
- Advanced search interfaces and analytics dashboards

The front-end will provide customized interfaces based on user roles and permissions.

3.3 Application Layer (Back-end)

The Application Layer handles business logic, APIs, authentication, and system processing services.

Technologies

- Node.js

Responsibilities

- Authentication and authorization
- API management
- Metadata processing
- File management
- Search services
- Payment processing
- Notification services and activity logging

The back-end architecture will be modular and scalable to support future institutional expansion.

3.4 Data Layer

The Data Layer manages structured and unstructured institutional datasets.

Technologies

- PostgreSQL
- Elasticsearch
- File Storage System

Responsibilities

- Data storage and retrieval
- Metadata indexing
- GIS data management
- Search optimization
- Backup and recovery
- Audit logging

Technology Stack

Component	Technology
Frontend	React.js
Backend	Node.js
database	PostgreSQL
Search Engine	Search Engine
Authentication	JWT / OAuth
Payment Integration	Tellbirr/ CBE
Hosting Environment	Linux Server / Cloud
Version Control	Git/GitHub

5. Database Schema and Structure

5.1 Database Design Approach

The proposed platform will use PostgreSQL as the primary relational database management system because it supports:

- Complex relational structures
- GIS spatial data support
- Advanced indexing
- High scalability
- Secure transaction processing

The database architecture will support geological reports, GIS layers, metadata records, payment transactions, user management, and audit logging.

5.2 Core Database Tables

Table	Purpose
users	User accounts
Roles	Permissions
Documents	Report metadata
Categories	Resource classification
gis_data	GIS datasets
uploads	Upload tracking
payments	Payment records
downloads	Download history
audit_logs	Activity logs
notifications	System alerts

5.3 Database Relationships

Users and Roles: Each user account will be linked to specific roles and permissions.

Documents and Categories: Reports and resources will be associated with metadata classifications and categories.

GIS and Metadata Integration: Spatial datasets will be linked to metadata records and geographic attributes.

Payments and Downloads: Payment records will be associated with purchased reports and download histories.

6. Geological Metadata Structure

The proposed system requires a flexible metadata structure capable of supporting geological reports, GIS datasets, laboratory records, and institutional resources.

The metadata framework is designed based on the existing Geological Institute Online Cataloging Service while improving searchability, accessibility, and data organization.

6.1 Core Metadata Fields

Metadata Field	Description
Title	Report title
Author	Responsible author/organization
Subject	Geological classification
Location	Study area
Published Year	Publication year
Report Number	Unique identifier
Collection	Resource category
Material Type	Report, GIS, Map, etc.
Abstract	Short summary
Keywords	Search tags
Format	PDF, DOCX, SHP
Organization	Source department
Language	Resource language
Pages	Total pages
Upload Date	Upload date
Access Level	Public/Restricted/Premium
Price	Premium report cost

6.2 Metadata Management Features

The system should support:

- Metadata creation and editing
- Mandatory metadata validation
- Metadata indexing
- GIS-linked metadata

- Advanced filtering
- Bulk metadata import/export
- Automatic metadata extraction
- Metadata standardization

6.3 Metadata Search Capabilities

The metadata architecture will support:

- Search by title
- Search by author
- Search by subject
- Search by location
- Search by published year
- Search by ID

7. Search and Indexing Architecture

7.1 Elasticsearch Integration

Elasticsearch will be integrated with PostgreSQL to provide:

- Fast metadata search
- PDF content search
- Advanced filtering
- High-speed retrieval performance

7.2 Search Workflow

Process Flow

1. User submits search query.
2. Query sent to Elasticsearch.
3. Indexed metadata and documents searched.
4. Permission filtering applied.
5. Results displayed through frontend dashboard.

8. System Workflow Processes

8.1 Data Ingestion Workflow

The data ingestion workflow defines the process for uploading, validating, storing, and publishing geological reports and institutional datasets within the system.

Workflow Process

1. Admin uploads file through the system upload interface.
2. Metadata is entered through the upload form.
3. System validates metadata fields and file type.
4. File is securely stored in the digital repository.
5. Metadata is stored in PostgreSQL database.
6. Elasticsearch indexes metadata and document contents.
7. Data Validator reviews and verifies the submission.
8. Approved content becomes searchable and accessible based on access permissions.

Workflow Objectives

- Ensure metadata quality and consistency
- Support secure file management
- Improve document searchability
- Enable centralized digital preservation
- Maintain controlled publication workflow

8.2 Authentication Workflow

The authentication workflow manages secure user login, identity verification, and permission control.

Workflow Process

1. User submits login credentials.
2. Back-end validates user account information.
3. JWT authentication token is generated.
4. User role and access permissions are verified.
5. Access is granted according to assigned privileges.

Authentication Features

- Secure login and session management
- Password encryption
- JWT-based authentication

- Role-Based Access Control (RBAC)
- Activity logging and monitoring

Security Objectives

- Prevent unauthorized access
- Protect sensitive geological information
- Support secure institutional operations
- Control user-level permissions

8.3 Payment and Download Workflow

The payment and download workflow supports secure online purchasing and controlled access to premium geological reports.

Workflow Process

1. User views report abstract and metadata information.
2. User selects premium report for purchase.
3. Payment request is sent to Telebirr or Chapa payment gateway.
4. Payment verification is completed successfully.
5. Download permission is automatically activated.
6. Transaction details are recorded in the database.

Payment System Features

- Secure online payment integration
- Premium report purchasing
- Download authorization management
- Transaction tracking and reporting
- Invoice and payment history management

Workflow Objectives

- Enable digital monetization of geological resources
- Support secure payment processing
- Improve accessibility to premium reports
- Maintain transparent transaction records

9. User Roles and Access Structure

The proposed system will implement Role-Based Access Control (RBAC) to manage permissions and secure institutional resources.

9.1 Public User

Public users are external users who access publicly available resources and services.

Permissions and Services

- Create login account using email address
- Login using username and password
- Browse metadata records
- Search public resources
- View report abstracts
- Purchase premium reports
- Download approved public files

Access Level: Limited access to publicly available resources and purchased materials only.

9.2 Registered Researcher

Registered researchers are authorized users with access to research-related resources and assigned datasets.

Permissions and Services

- Access assigned datasets
- Download purchased reports
- Access research materials
- View personal activity history
- Access specialized geological resources

Access Level: Controlled access based on institutional permissions and assigned datasets.

9.3 Data Validator

Data Validators are institutional staff members responsible for maintaining metadata quality and validating uploaded resources.

Permissions and Services

- Review uploaded resources

- Validate metadata quality
- Approve or reject submissions
- Maintain metadata standards
- Monitor document classifications

Responsibilities

- Ensure metadata consistency
- Verify uploaded content quality
- Support institutional data governance

9.4 System Administrator

System Administrators are responsible for managing operational and technical aspects of the platform.

Permissions and Services

- Manage users and permissions
- Configure payment systems
- Monitor system security
- Manage backups and recovery
- Configure system settings
- Monitor system performance

9.5 Super Administrator

The Super Administrator has the highest level of system authority and full access to all platform operations and database management services.

Permissions and Services

- Add, edit, and delete data from database
- View all datasets and system records
- Give access permissions for other users
- Manage administrator accounts
- Control system-wide configurations
- Manage user roles and privileges

Monitor all platform activities

10. Security Architecture

10.1 Authentication and Authorization

Security mechanisms include:

- JWT authentication
- Role-Based Access Control (RBAC)
- Password encryption
- Session management

10.2 Data Security Measures

The platform will implement:

- HTTPS encrypted communication
- Secure API communication
- File access restrictions
- Audit logging
- Backup and recovery systems

10.3 Backup and Disaster Recovery

The system will support:

- Automated database backup
- Incremental backup strategies
- Disaster recovery planning
- Cloud-ready redundancy mechanisms

11. Testing Strategy

Testing Type	Purpose
Unit Testing	Validate individual components
Integration Testing	Verify module interaction
Security Testing	Identify vulnerabilities
Performance Testing	Evaluate system efficiency
User Acceptance Testing	Validate business requirements

Testing activities will ensure reliability, security, and operational stability before deployment.

12. Deployment Strategy

Deployment Environment

- Linux-based hosting environment
- Local server infrastructure
- Docker container support
- Scalable server architecture

Deployment Activities

- Production server setup
- Database migration
- Security configuration
- SSL and domain setup
- Monitoring system activation

13. Sustainability and Maintenance

To ensure long-term sustainability, the project will include:

- Technical documentation
- Knowledge transfer sessions
- User training programs
- Six-month post-launch maintenance
- Continuous monitoring and issue resolution

The maintenance period will support operational stability and institutional transition.

14. Conclusion

The proposed methodology, system architecture, database structure, and metadata framework provide a scalable and secure foundation for implementing the Geological Information Management System for the Ethiopian Geological Institute.

The integration of React.js, Node.js, PostgreSQL, Elasticsearch, and GIS technologies will provide a modern and high-performance platform capable of supporting centralized data

management, advanced search services, digital preservation, secure online access, analytics, and institutional decision-making.

The proposed Agile development methodology and implementation framework will ensure successful system delivery, continuous stakeholder participation, and long-term sustainably.